

- **Technical solutions for scalability:** Techniques such as sharding (dividing a database into smaller, more manageable pieces) and load balancing (distributing workloads evenly across resources) are critical for maintaining performance.

## Data backup and disaster recovery

- **Implementing robust backup solutions:** Regular backups are essential to protect against data loss. Cloud platforms offer various backup services, but organisations must ensure they are configured correctly.
- **Strategies for effective disaster recovery:** A comprehensive disaster recovery plan, including regular testing and updating, ensures that data can be quickly restored in case of an outage or failure.

## Best practices in data management

Managing data effectively in the cloud is crucial for security, compliance, and getting the most out of your information. Here are some top best practices to consider.

## Security

### Encryption throughout the journey:

Implement encryption for your data at rest (stored in the cloud) and in transit (moving between locations). This makes data unreadable without a decryption key, safeguarding it from unauthorised access. Utilise cloud provider encryption solutions and understand how to manage your encryption keys effectively.

### Data governance framework:

Establish a robust data governance framework that dictates how data is handled in the cloud. This includes classifying data based on sensitivity (e.g., financial data requiring stricter controls), setting access controls (who can view or modify data), and outlining data retention policies (how long data is stored).

### Identity and access management

**(IAM):** Implement a granular IAM system to control access to your cloud data. This ensures only authorised users have access to specific datasets based on their roles and permissions. Regularly review and update access controls to minimise the risk of unauthorised access.

## Data optimisation and life cycle management

### Data life cycle management:

Implement policies to automate tasks associated with your data throughout its life cycle. This could include automatic archiving of inactive data to cheaper storage tiers, deletion of obsolete data based on retention policies, and migration of data between cloud environments for disaster recovery or cost optimisation purposes.

### Data cataloguing and

**classification:** Create a comprehensive data catalogue to track your data assets across the cloud and on-premises systems. This catalogue should include metadata (data about your data) such as data type, location, sensitivity, and ownership. A well-maintained data catalogue helps you understand what

information you have, where it resides, and who is responsible for it.

**Tiered storage:** Utilise a tiered storage strategy to optimise storage costs. Cloud providers offer various storage options with different price points based on access frequency. Store frequently accessed data in high-performance tiers and less frequently accessed data in lower-cost tiers.

### Data backup and disaster recovery:

Establish a data backup and disaster recovery plan to ensure your data is recoverable in case of outages or disruptions. Regularly test your backup and recovery procedures to ensure they function as intended.

### User training and education:

Train your users on cloud data security best practices. This includes educating them on data classification, password hygiene, and how to identify and report suspicious activity.

**Compliance:** Be aware of any relevant industry regulations or compliance requirements that apply to your data. Cloud providers offer tools and services to help you meet compliance standards, but it's your responsibility to understand your obligations. **END** 🐧

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## By: Dr Magesh Kasthuri and Dr Anand Nayyar

**Dr Magesh Kasthuri** is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

**Dr Anand Nayyar** is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning, and cyber security.